INCH-POUND
MIL-PRF-39012/93B
25 January 1995
SUPERSEDING
MIL-C-39012/93A
24 September 1986

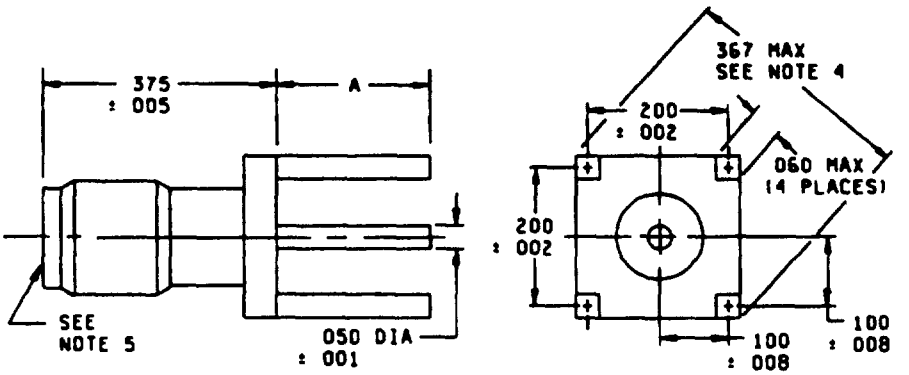
NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

PERFORMANCE SPECIFICATION

CONNECTORS, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES SMA (UNCABLED),FEMALE,PRINTED CIRCUIT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-39012



Dash number (x = shell material)	Dimension A inches (mm in parentheses)	Maximum overall length inches (mm in parentheses)
X001	155 (3 94) ± 010 (0 25)	545 (13 64)
X002	125 (3 18) ± 010 (0 25)	515 (13 08)
X003	093 (2 36) ± 010 (0 25)	483 (12 27

Inches	mm	Inches	mm	Inches	mm	Inches	mm
001	0 02	008	0 20	100	2 54	375	9 53
002	0 05	050	1 27	200	5 08		
005	0 13	060	1 52	367	9 32		

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 All undimensioned pictoral configurations are for reference purposes only
4 Configuration optional
5 Series SMA socket contact interface in accordance with MIL-STD-348
6 All corrosion resistant steel bodied connectors shall be gold plated in accordance with MIL-G-45204 type II class I

FIGURE 1 General configuration

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

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ENGINEERING DATA

Nominal impedance 50 ohms

Frequency range 0 to 18 GHz

Voltage rating 335 volts rms maximum at sea level 85 volts rms maximum at 70 000 feet

Temperature rating -65°C to +105°C

REQUIREMENTS

Dimensions and configuration See figure 1 and MIL-STD-348

Force to engage and disengage

Longitudinal force Not applicable

Torque 2 inch-pounds maximum

Coupling proof torque Not applicable

Inspection conditions For each test of threaded coupling connector where the test is performed on mated pairs the pairs shall be torqued to 7 to 10 inch-pounds

Mating characteristics See MIL-STD-348 for dimensions

Center contact (female)

Oversize test pin 0375 + 0001/- 0000 inch

Test pin finish 16 microinches

Insertion depth 030/ 045 inch

Number of insertions 3

Insertion force test

Steel test pin diameter 0370 + 0001/- 0000 inch

Insertion depth 050/ 075 inch

Test pin finish 16 microinches

Insertion force 2 pounds maximum

Withdrawal force test

Steel test pin diameter 0355 + 0000/- 0001 inch

Insertion depth 050/ 075 inch

Test pin finish 16 microinches

Withdrawal force 1 ounce minimum

Hermetic seal Not applicable

Leakage (pressurized connectors) Not applicable

Insulation resistance Method 302 test condition B MIL-STD-202 5 000 megohms minimum

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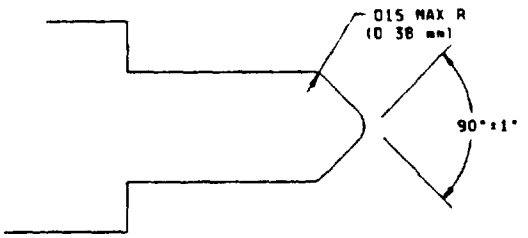


FIGURE 2 Test pin data

Center contact retention

Axial force 6 0 pounds minimum

Radial torque 4 inch-ounces minimum

Solderability 1/ Method 208 of MIL-STD-202

Corrosion (salt spray) Method 101 test condition B MIL-STD-202

Voltage standing wave ratio (VSWR) Not applicable

Connector durability Insertion and withdrawal force, 500 cycles minimum at 12 cycles per minute maximum The connector shall meet mating characteristics and force to engage and disengage requirements

Contact resistance (in milliohms maximum)

	Initial	After environment
Center contact	3 0	4 0
Outer contact	2 0	Not applicable

Dielectric withstanding voltage at sea level Method 301 of MIL-STD-202 1 000 volts rms

Vibration high frequency Method 204 test condition D MIL-STD-202

Shock (specified pulse) Method 213, test condition I MIL-STD-202

Thermal shock Method 107, test condition B MIL-STD-202 except test high temperature shall be +125°C

Moisture resistance Method 106 of MIL-STD-202 No measurements at high humidity Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity

1/ For quality conformance inspection the test shall be performed in group B following the insulation resistance test

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Corona level

Voltage 250 volts minimum

Altitude 70 000 feet

RF high potential withstanding voltage

Voltage and frequency 670 volts rms at 5 MHz

Leakage current Not applicable

Cable retention force Not applicable

Coupling mechanism retention force Not applicable

RF leakage Not applicable

Insertion loss Not applicable

Group qualification See table I

Part or Identifying Number (PIN) M39012/93 (dash number from figure 1) (see note below)

NOTE The center conductor terminal and the mounting legs may be tinned by the manufacturer if requested by an OEM The type of tinning and any related tests may be specified by the OEM The tinning operation shall not cause the connectors to otherwise fail these specification requirements The parts are to be marked with the original PIN s in effect prior to the tinning operation Tinned parts are for the OEM s immediate use only and should not be stocked and issued by the Government for replacement purposes

TABLE I Group qualification and retention testing

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	M39012/93-X001 M39012/93-X002 M39012/93-X003	M39012/93-X001 M39012/93-X002 M39012/93-X003

X = Material

NOTE If a connector manufacturer produces a connector which meets all the requirements for two or more connector types (within the same series) the manufacturer may receive qualification approval for the two or more connector types by qualifying the one connector It is not necessary that such connectors be in the same group Each connector however must be marked with its own appropriate PIN

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Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - CR
Navy EC
Air Force - 85

Review activities
Army - AM AT ME MI
Navy AS MC OS SH
Air Force 19 99

Preparing activity
DLA - ES

(Project 5935-3911-04)

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